

Donald V. Divicay Jr.

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510 - A Arco Alley St., Labangon, Cebu City

TECHNICAL SKILLS

Embedded Systems & Firmware

- Embedded C/C++, Zephyr RTOS, FreeRTOS
- Driver Development (I2C, UART, SPI, BLE, RS485 Modbus)
- Sensor Integration & Data Acquisition
- Firmware Architecture & Modular Design

Hardware & Tools

- ESP32, ESP8266, Nordic nRF
- PCB Design (KiCad), Debugging Tools

Software

- Flutter, React, Node.js, PHP, Javascript
- Git, CMake

LANGUAGES

C, C++, C#, Python, Java, JavaScript, Dart, Verilog, SQL

COMPUTER ENGINEER

Computer Engineer with experience in firmware development, multi-sensor integration, and low-level driver design using Zephyr RTOS and ESP-IDF. Skilled in building reliable systems, debugging complex hardware-software interactions, and ensuring data integrity across communication protocols.

Highly trainable and strong problem-solver with a reliability-focused mindset, experienced in system bring-up, performance optimization, and maintaining stability in real-world environments.

PROFESSIONAL EXPERIENCE

Xeleqt Innovations Incorporated

Jan - April 2026

Embedded Systems Engineer Intern

Designed and implemented modular multi-sensor firmware architectures to improve system scalability and maintainability, while developing custom drivers for I2C, UART, and SPI sensors to enable reliable real-time data acquisition. Built BLE scanning and parsing pipelines for wireless communication and implemented RS485 Modbus drivers with CRC validation and structured data handling. Performed hardware debugging, signal validation, and firmware bring-up to ensure stable and efficient system operation.

Department of Science and Technology

June - Aug 2025

Software Developer Intern

Developed and implemented web widgets for government websites using PHP and JavaScript, focusing on improving usability, functionality, and maintainability. Collaborated with a development team to integrate features, troubleshoot issues, and ensure reliable performance across different systems and user environments.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Engineering

Cebu Technological University - Main Campus
2022-2026

Department of Science and Technology

Junior-Level Science Scholarship - Merit
2024-2026

IIEE - Engineering Research Colloquia 2026

Overall Best Paper, Best Paper in Circuits, Sensors and Devices

nRF Connect SDK Intermediate (2026)

Nordic Semiconductor

Bluetooth Low Energy Fundamentals (2026)

Nordic Semiconductor

TESDA Certifications (Data Analysis, IoT, Java, CSS NCII)

Call Center Academy

PROJECTS

Non-Invasive Fish Freshness Assessment via Intelligent Bio-Capacitance Sensing

Developed a non-invasive fish freshness assessment system using bio-capacitance sensing for real-time, accurate classification.

ZenTask

Developed a full-stack task management web application using React, Node.js, and MongoDB with authentication and efficient task tracking.

Multi-Sensor Firmware Platform

Developed a modular multi-sensor firmware platform using Zephyr RTOS, integrating multiple sensors handling ambient parameters with centralized data management and real-time monitoring.